

Joining Multiple Tables

Till now we have seen one to one relationships where one table was joined with the values of another table. There are also possibilities when for each row in the left table there is more than one matching value. Such relationships are termed as one-to-many relationships. Similarly, there can be two tables to have many-to-many relationships being termed as many-to-many relationships. So, there are two ways a table can have a relationship with another table.

ONE TO ONE: A single record or row in one table has a relation to one row in another table.

ONE TO MANY: A single record or row in one table has relation to many rows in another table.

MANY TO MANY: Multiple records in one table have relation to many records in another table.

For instance, one has several relationships within their family. You have one mother, so you and mother are related. Though she has several children. Then you and your sibling are related in one way. Then you get married and have a spouse in one way. Similarly, tables inside a database or multiple databases can also be associated to get specific results.

ONE TO MANY:

In these types of relations, we have more than one corresponding rows on the right side for each row in the left table respectively. These relationships are called as one-to-many relationships. Here we are taking two tablets. The first table has an id, first_name, and last_name.

ID	FIRST_NAME	LAST_NAME
1	Robin	Jackman
2	Taylor	Edward
3	Vivian	Dickens
4	Harry	Clifford

The second table has an id, employee_id, type, and no respectively. Notice there are some duplicates for the employee_id with a one-to-many relationship as employees might have more than one telephone number.

ID	EMPLOYEE_ID	TYPE	NO
1	1	mobile	245-249697